



BRAC UNIVERSITY
Department of Computer Science & Engineering
FINAL EXAMINATION, Fall 2025
CSE 350: Digital Electronics and Pulse Techniques



Total Marks: 60 (20 + 20 + 20)

Time: 105 Minutes

- Total no of questions are **three (3)**. Answer **them all**.
- Figure in bracket [] next to each question indicates marks for that question.
- Write your set number on top of your answer sheet.
- Use $V_{BE_{sat}} = 0.8 V$, $V_{CE_{sat}} = 0.2 V$, $V_D = 0.7 V$, logic HIGH input = source voltage, logic LOW input = 0 V, if not stated otherwise.

Name:	ID:	Section:
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■ **Question 1** [CO2, CO1] [20 marks]

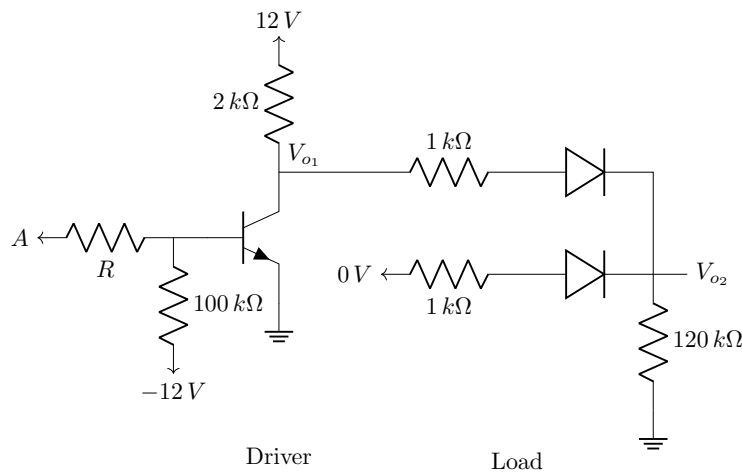


Figure 1: RTL Inverter driver and OR gate load circuit

For the above circuit, $\beta_F = 25$, $V_{OH} = 10 V$, $V_{OL} = 0.2 V$. Logic HIGH = 12 V, logic LOW = 0.2 V.

- For the driver, a consumer specifies that the power dissipation when $A = 0$ must be **1.24 mW** with [6]
no load connected. Find the value of the resistor R .
- Calculate the maximum fanout for the driver and the load. Find also the maximum power dissipation [8]
of the entire circuit.

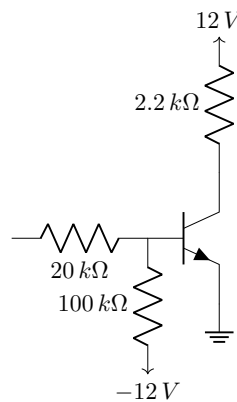


Figure 2: New load

- The load is **replaced by the new load** shown in figure 2. Find the noise margin of the circuit. [6]

■ Question 2 [CO2, CO1] [20 marks]

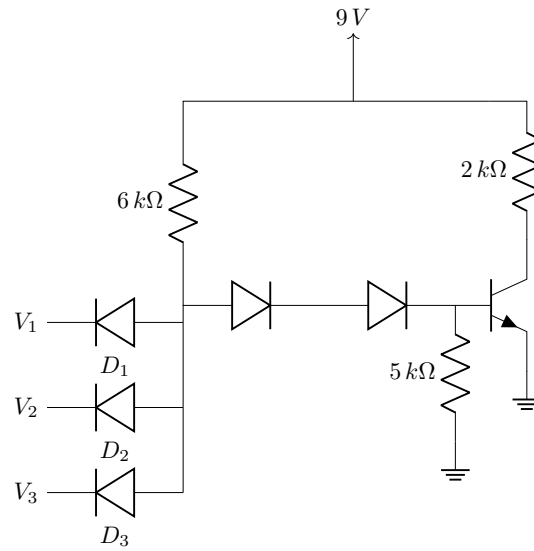


Figure 3: A 3-input DTL NAND Gate

For the above circuit, $V_{OH} = 9V$, $V_{OL} = 0.2V$, $\beta_F = 25$. If this circuit is used as both the driver and the loads—

- (a) Find the maximum fanout of the driver. [8]
- (b) What is the **minimum** current through D_1 , D_2 , and D_3 , when the driver's output is **high**? [2]

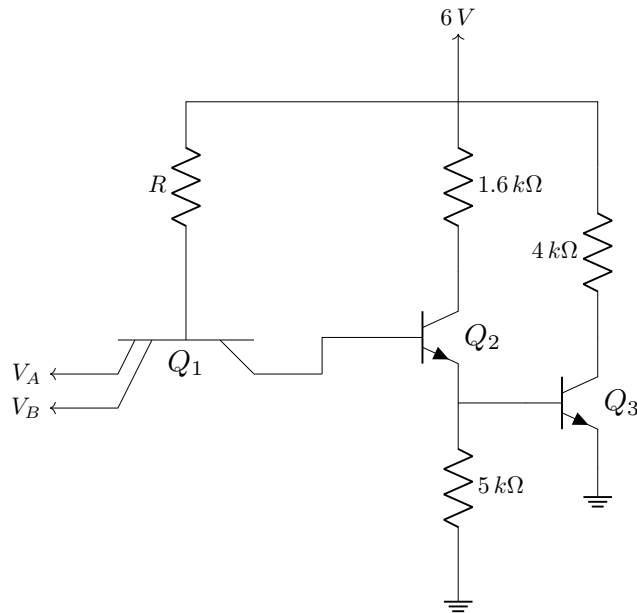


Figure 4: A 2-input TTL NAND Gate

For the above circuit, $\beta_R = 0.2$. If the emitter current of Q_3 is $5.71mA$ for a certain input logic condition—

- (c) Find the value of R . [10]

■ Question 3 [CO1] [20 marks]

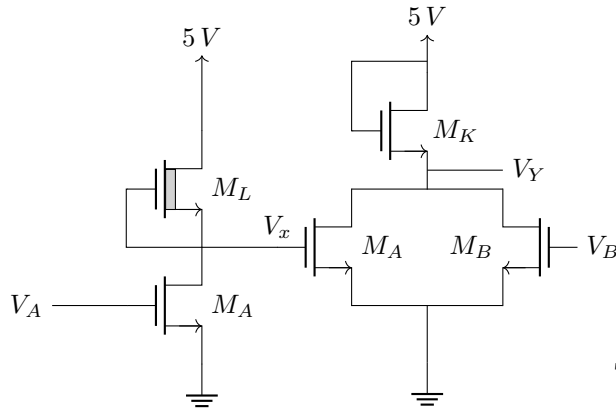


Figure 5

	$K_n \left(\frac{mA}{V^2} \right)$	$V_{TN} (V)$
M_L	0.5	-0.4
M_A	0.65	0.5
M_B	0.75	0.6
M_K	0.3	1

Table 1: MOSFET parameters

Triode Mode: $I_{DS} = K_n[2(V_{GS} - V_{TN})V_{DS} - V_{DS}^2]$

Saturation Mode: $I_{DS} = K_n(V_{GS} - V_{TN})^2$

- Find V_x & V_Y when both the inputs V_A & V_B are **high**. [8]
- Find the power dissipated in the **load** MOSFETs (M_L & M_K) for the case in (a). [5]
- What logic function does the above circuit perform (consider output V_Y and inputs V_A, V_B)? Implement this function using CMOS Logic. [7]